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Q: What are Neural Networks and why are they important in AI?

Ans A Neural Network, or Artificial Neural Network (ANN), is a computational model inspired by the way biological neurons in the human brain are connected and process information. It consists of many small processing units called artificials neurons, connected to one another through weighted connections.

A neural network usually contains three types of layers:

- Input layer – receives the input data.
- Hidden layer – process the data & learn patterns.
- Output layer – produces the final prediction or result.

Each connection has a weight that represents its importance. During training, the network is given examples and adjust these weights so that its predictions become more accurate. Therefore, unlike a traditional program, in which every rule is explicitly written by the programmer, a neural network can learn patterns from data.

For example, instead of writing thousands of rules describing what a cat looks like, a neural network can be trained on many labeled images of cats and other objects. It gradually learns useful features & then classify new image.

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Importance of Neural Networks in AI:-

Neural Networks are important because they allow AI systems to:

1. Learn from large amounts of data instead of depending only on manually written rules.
2. Recognize difficult patterns that are difficult to describe using ordinary programming.
3. Handle tasks such as image recognition, speech recognition, natural language processing, prediction and decision making.
4. Form the foundation of deep learning, where networks contain many layers.
5. Power many modern AI systems, including (LLMs) and other generative AI applications.

Q What is Prompt Engineering?

Ans Prompt Engineering is the process of designing and improving the instructions given to a Generative AI or LLM so that it produces a more accurate, relevant and useful response.

A prompt is the input or instruction given to an AI model. It may contain a

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question, task, context, constraints, or a required output format.

Good prompt engineering may include:

- giving clear and specific instructions;
- providing necessary context;
- defining the desired role or perspective;
- specifying constraints such as length or scope;
- providing examples of the expected result;
- specifying the required output format;
- refining the prompt when the first response is not satisfactory.

For example, a weak prompt might be:

"Explain AI".

A better engineered prompt would be:

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Chapter #01

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1-1 Define in your own words : a) intelligence b) artificial intelligence, c) agent. d) rationality e) logical reasoning.

a) Intelligence: It is the ability to perceive and understand information, learn from experience, reason, solve problems, adapt to new situations, and make effective decisions to achieve goals.

b) Artificial Intelligence: It is the field of science and engineering concerned with creating machines or computer systems capable of performing tasks that require intelligente behaviour, such as learning, reasoning, problem solving, perception and decision making.

c) Agent: An agent is an entity that perceives its environment and take actions within that environment. An intelligent agent uses the available information to choose actions that help it achieve its objectives.

d) Rationality: Rationality is the ability to choose the action that is expected to produce the best outcome, based on the available information, knowledge and goals of the agent.

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e) **Logical Reasoning**: It is the process of using known facts, rules or premises to derive valid conclusions in a systematic way. It enables an intelligent system to use existing knowledge to decide what follows from it.

1.2 Read Turing's original paper on AI (Turing 1950). In the paper, he discusses several objections to his proposed enterprise and his test for intelligence. Which objections still carry weight? Are his refutations valid? Can you think of new objections arising from developments since he wrote the paper? In the paper, he predicts that, by the year 2000, a computer will have 30% chance of passing a 5 minute Turning Test with an unskilled interrogator. What chance do you think a computer would have today? In another 50 years?

Ans Alan Turing proposed the Turing Test as a practical way of judging intelligent behaviour. Instead of asking whether a machine can literally "think", the test asks whether a human interrogator can distinguish a machine from a human through conversation.

Some objections discussed by Turing have become much weaker. For example, the claim the machines can only do exactly what programmers tell them is challenged by modern learning systems, which can learn patterns and produce results that

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were not explicitly programmed.

However, some objections still carry weight. A machine may produce human-like answers without necessarily having real understanding, consciousness or common-sense knowledge. Modern systems can also make convincing but incorrect statements, showing that fluent conversation alone is not a complete measure of intelligence.

New concerns include whether a system is truly reasoning or mainly reproducing patterns learned from large amounts of data.

Today, advanced AI systems would have a high chance of fooling an unskilled interrogator in a short, text-only conversation. In another 50 years, this probability may become even higher, although passing the Turing Test still would not by itself prove complete human-level intelligence.

1.3 Are reflex actions (such as flinching from a hot stove) rational? Are they intelligent?

Ans A reflex action can be rational if it produces the best expected result in a particular situation. For example quickly pulling a hand away from a hot stove prevents injury, so the action is rational

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even though no careful reasoning takes place.

However, a reflex action by itself is not necessarily intelligent. Reflexes are usually automatic responses to specific conditions and may not involve learning, reasoning or planning.

Therefore, rational behaviour does not always require logical reasoning. A simple reflex can be rational when an immediate action is more useful than spending time thinking about it.

1.4 Suppose we extend Evan's Analogy program so that it can score 200 on a standard IQ test. Would we then have a program more intelligent than a human? Explain.

Ans No. A score of 200 on an IQ test would show that the program is extremely capable at the specific abilities measured by that test, but it would not prove that the program has greater overall intelligence than a human.

Intelligence include many abilities such as learning, reasoning, perception, communication, adaption, creativity and solving unfamiliar problems. A program ~~can~~ may perform exceptionally well in one narrow area while lacking abilities outside that area.

Therefore, intelligence should not be judged by a single test score. The program would demonstrate

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high performance on the IQ test, but this alone would not establish general human-level or superhuman intelligence.

1-5 The neural structure of the sea slug Aplysia has been widely studied (first by Nobel laureate Eric Kandel) because it has only about 20,000 neurons, most of them large and easily manipulated. Assuming that the cycle time for an Aplysia neuron is roughly the same as for a human neuron how does the computational power, in terms of memory updates per second, compare with the high-end computer in Fig 1.3?

Ans According to Fig 1.3, the cycle time of human neuron is approximately:

$$10^{-3} \text{ seconds}$$

Aplysia has approximately:

$$20,000 = 2 \times 10^4 \text{ neurons}$$

Therefore, its approximate memory updates per second are:

$$\frac{2 \times 10^4}{10^{-3}} = 2 \times 10^7$$

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So, Aplysia can perform approx:

2×10^7 memory updates/ second

The high-end computer in Fig 1.3 performs approx:

10^{14} memory updates/ second

Comparing them:

$$\frac{10^{14}}{2 \times 10^7} = 5 \times 10^6$$

∴

Therfore, the high-end computer performs roughly 5 million times more memory updates per second than Aplysia.

1.6 How could introspection—reporting on one's inner thoughts—be inaccurate? Could I be wrong about what I'm thinking? Discuss.

Ans Introspection means observing and reporting our own thoughts and mental processes. It can be inaccurate because a person does not always have complete access to how the mind actually produces a though or decision.

Our reports may be affected by memory errors, personal bias, emotions, unconscious mental processes, or incorrect interpretation of our own thoughts. For example, a person may know

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what decision they made but may incorrectly explain the real reason behind that decision.

Therefore, yes, we can sometimes be wrong about what or why we are thinking. This is one reason cognitive science does not rely only on introspection; it also uses psychological experiments and brain observations to study human thinking.

1.7 To what extent are the following computer systems instances of artificial intelligence:

- Supermarket bar code scanners.
- Web search engines.
- Voice-activated telephone menus.
- Internet routing algorithms that respond dynamically to the state of the network.

Ans Whether these systems are considered AI depends on how much perception, reasoning, learning, adaption, and intelligent decision making they perform.

1) A basic barcode scanner is not considered AI. It reads a predefined code and retrieves the corresponding information using fixed rules.

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2) Web search engines are strong examples of systems that use AI techniques. They analyze queries, rank information, understand language, and may learn from data to provide more relevant results.

3) Voice-activated telephone menus can involve AI because they use speech recognition to interpret spoken input. More advanced systems may also use natural language processing to understand what the user want.

4) Dynamic internet routing algorithms show some intelligent behaviour because they observe changing network conditions and adapt their routing decisions. However, traditional routing algorithms are usually considered conventional algorithms rather than full AI systems.

Thus, AI is not simply about automation; the degree of adaptation, learning, reasoning, and decision making matters.

1.8 Many of computational models of cognitive activities that have been proposed involve quite complex mathematical operations, such as convolving an image with a Gaussian or finding a minimum of the entropy function. Most humans (and certainly all animals) never learn this kind of mathematics at all, almost no one learns it before college, and almost no one can compute

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the convolution of a function with a Gaussian in their head. What sense does it make to say the "vision system" is doing this kind of mathematics, whereas the actual person has no idea how to do it?

Ans A person's brain can perform processes that can be described mathematically without the person consciously knowing the mathematics involved

When researchers say that the visual system performs an operation similar to convolution or optimization, they do not mean that the person consciously writes equations and solve them. Instead the mathematical operation is a model used to describe the underlying information processing performed by the nervous system.

For example, a person can recognize a face almost instantly without knowing the calculations involved in visual processing.

Therefore, there is a difference between performing a computational process internally and consciously understanding the mathematical description of that process.

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1.9 Why would evolution tend to result in systems that act rationally? What goals are such systems designed to achieve?

Ans Evolution tends to favour organisms whose behaviour helps them survive and reproduce successfully. An organism that chooses actions appropriate to its environment is more likely to survive, obtain resources, avoid danger, reproduce, and pass its gens to future generations.

Therefore, natural selection can gradually produce systems that behave in ways that appear rational, because ineffective behaviours are less likely to contribute to reproductive success.
The immediate goals of such systems may include:

- finding food and other resources;
- avoiding danger;
- protecting themselves and their offspring;
- finding mates;
- adapting to environmental conditions.

Ultimately, from an evolutionary perspective, these behaviours contribute to survival and reproductive success.

Evolution does not consciously design these goals; rather they merge through the process of natural selection.

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1.10 Is AI a science, or is it engineering? Or neither or both? Explain.

Ans Artificial Intelligence can be considered both a science and an engineering discipline.

- AI as a Science: AI studies the principles behind intelligence, including learning, reasoning, perception, decision making, knowledge representation and intelligent behaviour. Researchers develop theories, construct models, perform experiments, and evaluate results scientifically.

- AI as Engineering: AI also applies these principles to design and build intelligent systems that solve practical problems, such as autonomous systems, search engines, language-processing systems, robots, and decision-support systems.

Therefore, AI combines:

- Science → understanding intelligence and intelligent behaviour.

Engineering → building systems that demonstrate useful intelligent behaviour.

Hence, AI is best described as both a scientific and an engineering field.

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Q: Discuss the definitions of Artificial Intelligence with examples. Which one do you like most and why?

Ans: AI can be understood in 4 main ways, depending on whether we focus on thinking or acting, and whether we compare intelligence with humans or rational behaviour.

1. Thinking Humanly :-

This approach says that AI should try to think in way similar to the human mind. It studies how humans learn, remember, reason, and solve problems.

Example: An AI system designed to solve a problem using reasoning steps ~~to~~ similar to those used by a human student.

2. Acting Humanly :-

This approach considers a machine intelligent if it can behave like a human. The Turing Test is based on this idea: if a person communicates with a machine and cannot reliably tell it is a machine, the machine shows human-like intelligent behaviour.

Example: A chatbot that communicates naturally enough that its responses appear human.

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3. Thinking Rationally :-

This approach focuses on correct reasoning. The AI uses facts and rules of logic to reach valid conditions.

Example : If the system knows that all humans are mortal and Ali is a human, it can logically conclude the Ali is mortal.

4. Acting Rationally :-

This approaches says that AI should choose the best possible action according to the information it currently has and the goal it wants to achieve.

Example : A self-driving car detects an obstacle and chooses to break because braking is the safest action in that situation.

Which approach do I like most and why?

I like the Acting Rationally (Rational Agent) approach the most because an AI system does not need to copy

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humans exactly. What matters more is whether it can understand the situation, consider the available information, and choose an action that gives the best expected result.

Humans can also make mistakes, so simply copying human thinking or behavior is not always the best goal. The rational-agent approach is more practical because it focuses on making good decisions and achieving goals effectively.

Chapter # 02

Intelligent Agents

2.1 Suppose that the performance measure is concerned with just the first T time steps of the environment and ignores everything thereafter. Show that a rational agent's action may depend not just on the state of the environment but also on the time step it has reached.

Ans: A rational agent chooses the action that gives the best expected performance. If performance is measured only for the first T time steps, then the amount of time remaining can affect which action is best.

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